MIL-D-51083C
29 December 1967
SUPERSEDING
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4 November 1966

MILITARY SPECIFICATION

DETECTOR TICKET, ANTICHOLINESTERASE AGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of detector ticket designed to detect anticholinesterase agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-51189 - Substrate Solution and Dispenser.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

C5-77-808 - Belt, Detector Ticket, Assembly and Detail.
C5-77-827 - Ticket, Detector, Assembly and Detail.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6665

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

MUNSELL COLOR COMPANY, INC.

Munsell Book of Colors

(Application for copies of this book should be addressed to the Munsell Color Company, Inc., Baltimore, Maryland 21203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standard

D1193-56 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing C5-77-808 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the detector ticket shall conform to the specifications and drawings listed on Drawing C5-77-808 and on subsidiary drawings.

3.2 Manufacture and assembly. The detector ticket shall be manufactured to conform to Drawing C5-77-827. Each fiber glass disk shall be impregnated with 0.030 ± 0.005 milliliter (ml) of cholinesterase enzyme solution. The impregnated tickets shall be vacuum dried at 25° to 80°F

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under a pressure not to exceed 500 microns of mercury for no less than two hours. The tickets shall be stored over a desiccant at a temperature not to exceed 100°F until assembled into belts as specified on Drawing C5-77-808 (see 6.2). The assembly operation shall be conducted in an environment having a relative humidity of 40 percent maximum and a temperature of 70° ± 5°F.

3.3 Cholinesterase activity. Each disk shall give a cholinesterase activity of 0.8 to 1.6 units when tested as specified in 4.3.4.1.

3.4 Sensitivity. After aging for two weeks at 70°C, the detector ticket shall comply with the following:

3.4.1 Square end. The disk in the square end shall develop one of the colors in table I within 2-1/2 minutes after exposure to 0.10 microgram of GE when tested in accordance with 4.3.4.2(b)(1).

Table I. Munsell colors

Hue (Read clockwise)	Value	Chroma
5.0 P through 5.0 Y	7-8	2-8
5.0 P through 5.0 Y	9	0-8
5.0 Y through 5.0 P	9	0.5
5.0 R through 5.0 YR	7-8	10.0

3.4.2 Round end. The disk in the round end shall develop one of the colors in table II within 45 seconds when tested in accordance with 4.3.4.2(b)(2).

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Table II. Munsell colors

Hue (Read clockwise)	Value	Chroma
2.5 B through 5.0 PB	8	2-4
2.5 B	7	2
5.0 B through 5.0 PB	7	4-6
2.5 B	5-6	6
5.0 B through 10 B	5-6	4-6
2.5 PB	6	4-8
2.5 PB	5	4-10

3.5 pH of water extract. When determined in accordance with 4.3.4.3, the water extract of the impregnated disks shall have a pH between 8.0 and that of the water before extraction.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of detector tickets shall be produced in accordance with this specification for examination and test (see 4.2).

3.7 Workmanship. The detector tickets and belt assemblies shall be free from tears, rips, punctures, cracks, dirt, grease and oil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Preproduction.

4.2.1 Sample. A preproduction sample of four belts of detector tickets shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2).

4.2.2 Inspection.

4.2.2.1 For examination. Each detector ticket in the sample shall be examined for all the requirements of the applicable drawings and this specification.

4.2.2.2 For test. Sample detector tickets shall be tested in accordance with 4.3.4 and table III. However, no more than 5 tickets shall be taken from any one belt for any given test.

4.2.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination as specified in 4.2.2.1 and, when tested as specified in 4.2.2.2, shall comply with the acceptance numbers listed in table III to be acceptable.

4.3 Regular production.

4.3.1 Lotting. A lot of belts shall consist of the detector tickets impregnated with one batch of cholinesterase enzyme and produced by one manufacturer under the same conditions (see 6.3).

4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For tests. Sampling shall be conducted in accordance with table III. No more than 5 tickets shall be taken from one belt.

4.3.3 Inspection.

4.3.3.1 For examination. Sample detector tickets shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For tests. Sample detector tickets shall be tested in accordance with 4.3.4 and table III.

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Table III. Sampling plan

Test	No. of tickets	Acceptance No.
Cholinesterase activity*	10	1
Sensitivity*	20	2
pH of water extract	8	0

*Each disk from the tickets shall be tested separately.

4.3.3.3 Classification of defects.

(a) Belt, detector ticket, assembly and detail (Dwg C5-77-808).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing or incorrectly assembled
102	Component dimensions incorrect
103	Perforations incorrectly located
Minor:	AQL 2.5 percent defective
201	Number of tickets per belt incorrect
202	Marking incorrect
203	Color coding missing or incorrect
204	Workmanship (3.7)

4.3.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests, which shall be conducted as follows:

4.3.4.1 Cholinesterase activity.

(a) Reagents and equipment.

Sodium barbital solution. Dissolve 0.4124 gram (g) of sodium barbital in 30 ml of water and adjust the pH to 7.4 with 0.2N hydrochloric acid. Dilute to 100 ml.

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Magnesium chloride solution. Dissolve 4.07 g of magnesium chloride in sufficient water to make 100 ml.
Acetylcholine chloride solution. Dissolve 3.993 g of acetylcholine chloride in 8 ml of water and adjust the pH to 7.4 with normal sodium hydroxide. Dilute to 10 ml. Prepare prior to use.
Standard sodium hydroxide solution. 0.02N.
Titrator. pH Stat Type.
Magnetic stirrer.
Titration vessel. 8 to 10 ml capacity, controlled at 25° ± 1°C.

(b) Procedure. To the titration vessel, add 1.7 ml of water and one fiber glass disk. Break up the disk with a glass rod. Wash down the rod with 1.0 ml of sodium barbital solution and 1.0 ml of magnesium chloride solution, collecting the washings in the titration vessel. Fill the buret with the sodium hydroxide solution. Set the titrator at pH 7.4 and activate the instrument. When the solution has come to equilibrium as indicated by no addition of sodium hydroxide for approximately one minute, note the buret reading and then add 0.3 ml of the acetylcholine chloride solution. Immediately start a timer. At the end of five minutes, note the buret reading. Calculate as follows:

$$\text{Cholinesterase activity, units/disk} = \frac{1000\ AB}{C}$$

where: A = milliliters of sodium hydroxide used after addition of acetylcholine chloride,
B = normality of the sodium hydroxide solution, and
C = time interval in minutes.

4.3.4.2 Sensitivity.

(a) Apparatus and reagents.

Dilution Apparatus, Toxic Gas, Q5.
Tris buffer. Dissolve 0.607 g of tris-(hydroxymethyl)-aminomethane in 25 ml of water, add 28.7 ml of 0.1N hydrochloric acid, and dilute to 100 ml. Adjust the pH to 8.0.
Substrate solution. Prepare as directed in MIL-S-51189.

(b) Procedure. Cut the detector tickets in half and proceed as follows:

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(1) Square end. Wet the disk with two drops of tris buffer. Place the square end in a plastic pocket and massage the disk for 30 seconds. Remove the ticket from the pocket and expose the disk to 0.10 microgram of GB within 30 to 60 seconds at a flow rate of 1 liter/minute. Wet the disk with 3 drops of substrate solution. Replace the square end in the pocket and massage the disk for 30 seconds. Remove the ticket from the pocket and check for compliance with 3.4. Color under the retaining ring is permissible.

(2) Round end. Wet the disk with two drops of tris buffer. Place the round end in a clean plastic pocket and massage the disk for 30 seconds. Remove the ticket from the pocket, wet the disk with 3 drops of substrate solution, replace the round end in the pocket, and massage the disk for 30 seconds. After 15 additional seconds, remove the ticket from the pocket and check for compliance with 3.4.

4.3.4.3 pH of water extract. Remove 16 impregnated disks from the detector tickets and place them in a 30-ml beaker containing 5 ml of water (pH 6.0 to 7.0). Break up the disks with a glass rod and agitate the mixture vigorously for 5 minutes. Determine the pH using a pH meter.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The detector ticket belts shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Detector ticket belts, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The detector ticket covered by this specification is intended for use in chemical agent detector kits for the detection of anticholinesterase chemical agents.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of detector tickets per belt.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD-94. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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International interest (see section 6)

Preparing activity:

Army - MU (EA)

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